2022 IEEE International Students' Conference on Electrical, Electronics and Computer Science (SCEECS) | 978-1-6654-1418-0/22/$31.00 ©2022 IEEE | DOI: 10.1109/SCEECS54111.2022.9741058

# Design and Implementation of 4-bit and 8-bit KSA in 18nm FinFET Technology

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***Abstract*— The addition operation is required by any digital, analogue, or control system. The speed and accuracy with which the digital system operates is dependent on the functioning of adders. Adders are utilised in a variety of logical and arithmetic procedures. The goal of VLSI system design research is to improve the performance of digital systems, hence there is a strong incentive to examine and analyse adder architectures.**

**The purpose here is to understand the family of faster adders available, such as the Ripple Carry Adder, Carry Look Ahead Adder, Carry Select Adder, and so on.**

**This paper is about the Kogge Stone Adder in the 18nm FinFET model's technology node. It has a high PDP because it operates at a faster rate than its competitors. The adder is designed and built for bit lengths of 4 and 8, with random input data simulations used to evaluate its operations.**

**Criteria such as power, performance, area, gate count, and PDP are used to construct a comparison. Every block used to construct the adder is inspected, and the results are graphically shown. In order to properly examine the parameters, the acquired data is tabulated.**

***Index Terms*—Ripple Carry Adder (RCA), Carry Look Adder (CLA), Carry Increment Adder (CIA), Parallel Prefix Adder (PPA), Kogge-Stone Adder (KSA), FinFET.**

## I. Introduction

The need for electronic devices has increased dramatically as technology has advanced, necessitating the development of incredibly efficient VLSI logics.

DSPs, microprocessors, microcontrollers, and other processing units are made up of arithmetic units, which are the basic building blocks of digital systems [1].

A combinational circuit consists of many inputs, logic gates, and outputs. Combinational logic gates respond to the values at their inputs and produce the value of the output signal, transferring binary information from supplied input data to desired output data [1]. As a result, to design a combinational circuit, a truth table containing the output values for each combination of input variables may be employed.

In every digital system, digital adders are critical components since they have a significant influence on the overall performance that may be obtained. As a result, high-speed adders are quite popular. The speed of an adder may be increased by enhancing both the top-level design and the circuit implementation [3]. In general, addition computes the sum of 2 numbers, resulting in total and carry. All advanced adder circuits need the use of half (HA) and full adders (FA). Because adders have the greatest influence on the ALU, increasing speed while lowering power consumption has a significant impact on microprocessor speed and total energy consumption.

The carryout of each full adder is equal to the carry in of the next most important full adder in a ripple carry adder (RCA). Ripple carry adders are named that way because each carry bit ripples into the next step. When utilising RCA for binary addition, the carry chain is the most serious problem. As the breadth of the input operand widens, the carry chain length expands. The scenario grows significantly worse when the carry takes the longest route from the LSB to the MSB. To decrease RCA latency, we're thinking of employing a CLA adder. A carry look-ahead adder reduces propagation latency by incorporating more sophisticated circuitry than previous adder layouts [5]. The CLA Adder is a component of the Carry Increment Adder circuit, which also comprises incremental circuitry composed of half adders in a sequential order. When compared to other adders, the Carry select adder (CSA) has a quicker computational speed. On the other side, the circuit's complexity, increased power consumption, and large footprint are negatives.

The goal of providing a brief overview and drawbacks of basic adder architectures is to demonstrate that there is a compelling reason to seek out a better Adder architecture, to research, develop, and implement one.

## II. Parallel Prefix Adders

Parallel prefix adders are a sort of adders created from CLA adders that is suitable for larger words. To reduce latency, the PPA circuit adopts a tree-like architecture to ’0’(log2n), where

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the number of bits is represented by 'n'. The parallel-prefix adder approach can be used to overcome the latency of CLA adders [6]. The objective is to compute a small group of intermediate prefixes, then a large group of prefixes, and so on, until all of the carry bits have been computed. Parallel-prefix structures have been discovered to appeal to adders due to their logarithmic latency.

Three crucial phases are carried performed in parallel-prefix computing, as shown below:

1) Using the number of input bits, compute, generate, and propagate signals.
2) Prefix computation is the process of calculating all the carry sequences in parallel.
3) Calculate the sum of the provided inputs.

A variety of Parallel Prefix adders have been developed since the 1960s. The J. Sklansky Conditional Adder from 1960, the Kogge-Stone Adder from 1973, the Ladner-Fisher adder from 1980, and so on.

Parallel prefix adders are important because of their speed. This time efficiency is achieved through the use of a tree network known as a 'prefix-operation graph'.

## III. Kogge Stone Adder

The Kogge-Stone Adder, or KSA, is a parallel prefix variant of CLA invented by Kogge and Stone in 1973. It generates a carry in '0' (logn) time. It's also known as the faster adder, and it's employed in high-performance arithmetic circuits in industry.

KSA generates carry at a quicker pace by computing both the carry and the area at the same time. KSA offers a common architecture that allows integrating and utilising current technologies straightforward. Another advantage of KSA is its reduced fan-out [2].

The expression log2n is used to calculate KSA's latency. (n*log2n) – n + 1 is the area of the KSA, where n is the number of input bits. This adder architecture is shown to be more effective for higher-speed applications.

KSA's basic operation is broken down into three steps,which are listed below:

- Pre processing
- Carry Generation (Carry Look Ahead Network)
- Post processing

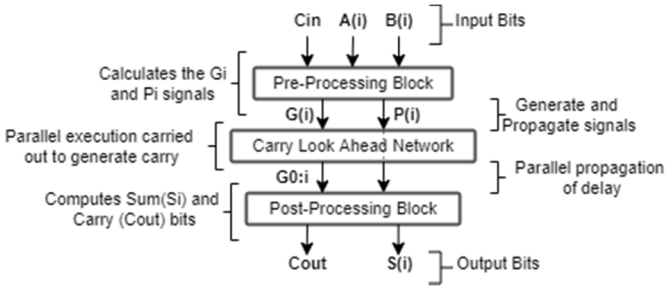



Fig. 1. Kogge Stone Adder: Stages of operation

### A. Architecture

*1) Pre processing:* This stage involves the computation of generate (Gi) and propagate (Pi) signals corresponding to the pair of bits in the inputs A and B.

*2) Carry generation:* This stage, also known as the "carry-look ahead network," entails calculating the carries for each bit (Ci). It uses group propagate and generate as intermediate signals, which are then employed to compute signals for the final Sum and Cout calculations. The carries are broken into separate parts after they have been computed in parallel. 2 AND gates and an OR gate are present in the carry operator.

*3) Post processing:* This is the last step in computing the sum of the inputs, and it is shared by all adders in this family (CLA). It entails sum and carry out bit computations.

The block diagram of an 8-bit Kogge-Stone Adder is shown in the next picture. Sub-blocks such as the PG (Propagate-Generate) block, FCO (Fundamental Carry Operator) block, PPC block, and XOR logic gate with linkages to build up the appropriate logic.

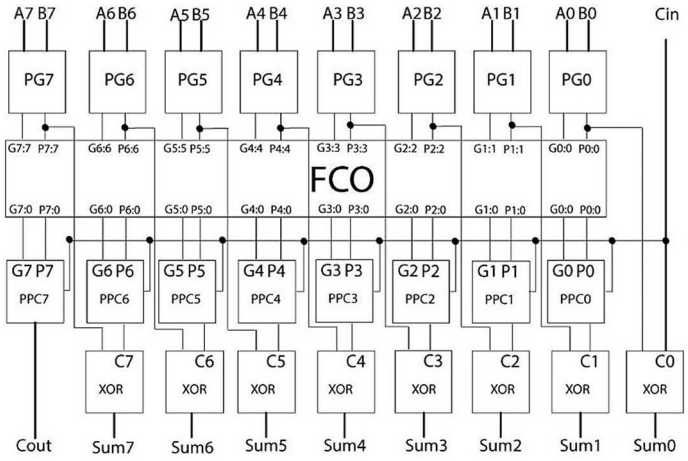



Fig. 2. Block diagram Architecture of an 8-bit KSA

### B. Standard Cells

Standard cells are pre-defined, pre-characterized cells that are utilised as fundamental building blocks in the ASIC (Application Specific Integrated Circuit) Design flow. These cells are of the same height, with varying widths, and can easily fit into a standard cell row. They may be used again and over again. A Boolean logic function is made up of a group of linked transistors known as Standard Cells.

Sub-blocks in this project work are built using the Standard Cells method.

### C. Sub blocks

The previously mentioned standard cells are instantiated in the sub blocks and linkages are built to generate the required simulation results. The schematics, symbols, and layouts for the sub blocks are shown here, along with their descriptions.

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*1)* ***Gen_Prop Block****:* Gen_Prop is a pre-processing block, which includes one XOR gate and one AND gate. This block computes the generate and propagate signals. The input bits to these gates are A and B. These signals are given by Logic equations as given below:

$$P_i = A_i \oplus B_i \tag{1}$$

$$G_i = A_i * B_i \tag{2}$$

TABLE I
TRUTH TABLE: GEN_PROP BLOCK

| Ai | Bi | Pi | Gi |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |

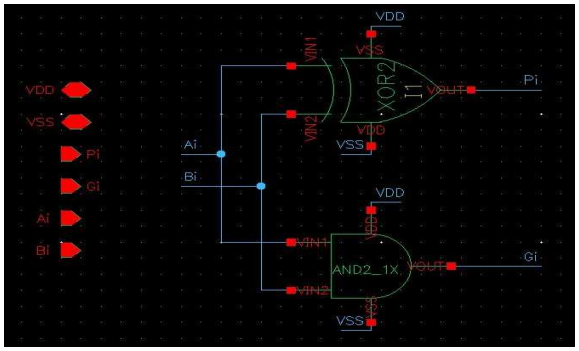



Fig. 3. Schematic: Gen_Prop Block

*2)* ***Gi_Pi Block****:* Gi Pi block is a Carry Look Ahead network, which includes two AND gates and one OR gate. This block computes the carries corresponding to each bit.The input bits are Gi, Gj, Pi, and Pj. It uses group propagate and generate as intermediate signals which are given by the logic expressions below:

$$P_{i:j} = P_{i:k+1} * P_{k:j} \tag{3}$$

$$G_{i:j} = G_{i:k+1} + (P_{i:k+1} * G_{k:j}) \tag{4}$$

TABLE II
TRUTH TABLE: GI_PI BLOCK

| Gi | Pj | Gj | Gij |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 |

TABLE III
TRUTH TABLE: GI_PI BLOCK

| Pi | Pj | Pij |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

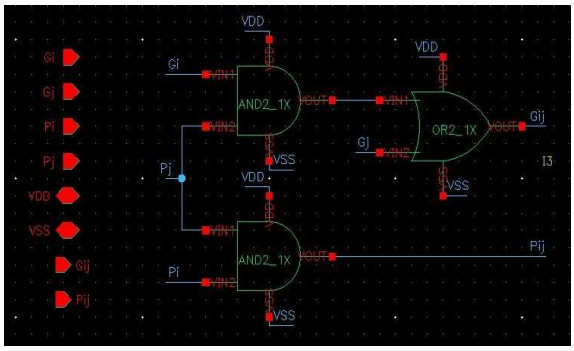



Fig. 4. Schematic: Gi_Pi Block

*3)* ***Gi_Block****:* Gi is a post processing block, which includes one AND gate and one OR gate. This block is the final step and is common to all adders of this family (carry look ahead). The input bits are Gi, Pj, and Gj. It involves computation of sum bits. Sum bits are computed by the logic equations as shown,

$$S_i = P_i \oplus C_{i-1} \tag{5}$$

$$C_{out} = G_i + (P_i * G_{i-1}) \tag{6}$$

TABLE IV
TRUTH TABLE: GI BLOCK

| Gi | Pj | Gj | G0j |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 |

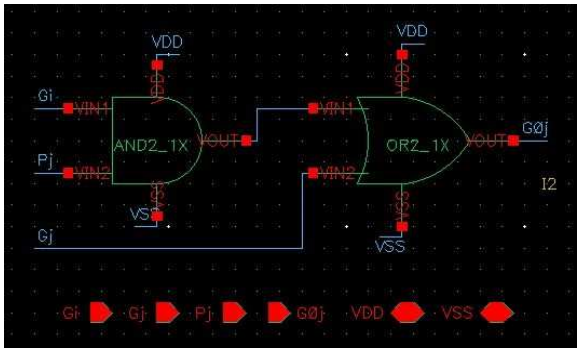



Fig. 5. Schematic: Gi Block

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### *D. 4-bit KSA*

The 4-Bit KSA Top-level schematic is implemented and is as shown in the figure 6. The corresponding layout is also created as shown in the figure 7.

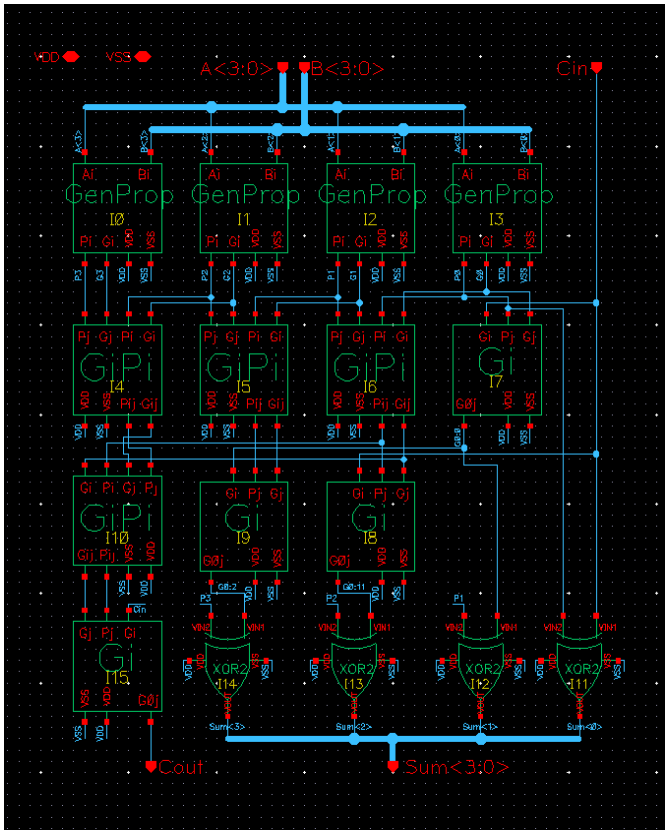



Fig. 6. Schematic: 4-bit KSA

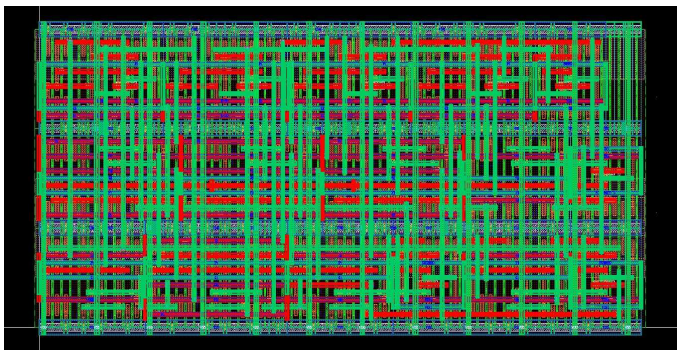


Fig. 7. Layout: 4-bit KSA

### *E. 8-bit KSA*

The 8-Bit KSA is implemented and is as shown in the figure 8. The corresponding layout is also created as shown in the figure 9.

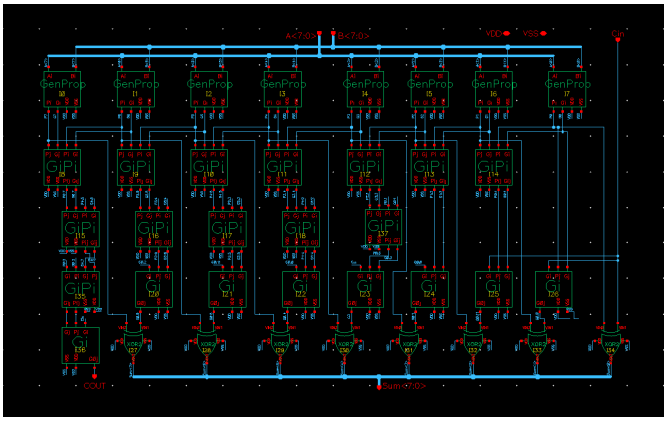


Fig. 8. Schematic: 8-bit KSA

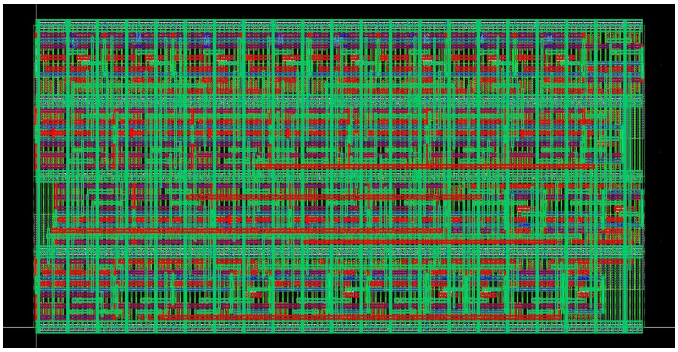


Fig. 9. Layout: 8-bit KSA

## IV. SIMULATIONS

To obtain the parameters, simulations are run on all of the blocks, and the data is tabulated and graphically represented such that the parameters may be properly compared and studied.

The functionality of the 4-bit KSA is shown in the fig 10.

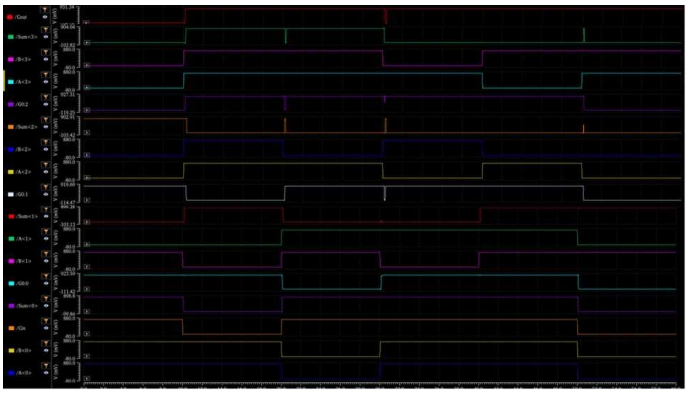


Fig. 10. Simulation Graph: 4-bit KSA

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The functionality of the 8-bit KSA is shown in the fig 10 and fig 11.

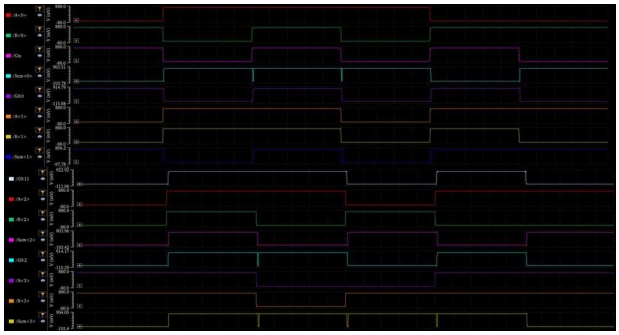


Fig. 11. Simulations: 8-bit KSA

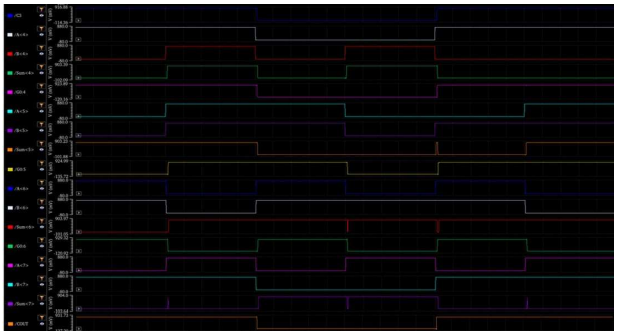


Fig. 12. Simulations: 8-bit KSA

## V. RESULTS

All the Primitive structures, sub blocks and the top-level implementation are verified for their functionalities, simulations are run on the 4-bit and 8-bit models.

The parameters are: Delay, Power, Power-Delay Product, Area utilization and the transistor/Gate Count.

The power consumption of each bit cell and sub block is calculated by altering the supply voltage, VDD, and tabulating theresults with regard to the supply voltage. As VDD increases, power consumption increases as well, while the delay of the same devices decreases.

The results of the delay calculation are achieved by altering one of the two input signals while keeping the other constant. The threshold voltage that is taken into account is half of the supply voltage, VDD. The technique is extended to the other input as well, and the maximum delay obtained is used in the later phases to evaluate the settings.

The Delay, Power and Power-Delay Product (PDP) at different frequencies of 4-bit KSA are tabulated in the tables 5-7.

TABLE V
4-BIT SIMULATIONS: AT 100MHZ

| VDD | Delay(ps) | Power at 100MHz(nW) | PDP(pJ) |
|---|---|---|---|
| 0.6 | 527.4690 | 1000.7800 | 52790.7 |
| 0.7 | 410.6770 | 1438.6900 | 59083.6 |
| 0.8 | 314.5874 | 1999.2300 | 68291.1 |
| 0.9 | 295.7002 | 2707.1200 | 80049.5 |
| 1 | 263.4064 | 3587.9000 | 94507.5 |

TABLE VI
4-BIT SIMULATIONS: AT 1.25GHZ

| VDD | Delay(ps) | Power at 1.25GHz(nW) | PDP (pJ) |
|---|---|---|---|
| 0.6 | 527.4690 | 8776.72 | 4629.6 |
| 0.7 | 410.6770 | 12154.5 | 4991.5 |
| 0.8 | 314.5874 | 16201.1 | 5534.09 |
| 0.9 | 295.7002 | 20995.2 | 6208.2 |
| 1 | 263.4064 | 26595.9 | 7005.3 |

TABLE VII
4-BIT SIMULATIONS: AT 2GHZ

| VDD | Delay(ps) | Power at 2GHz(nW) | PDP (pJ) |
|---|---|---|---|
| 0.6 | 527.4690 | 13831.3 | 7295.5 |
| 0.7 | 410.6770 | 19123.5 | 7853.5 |
| 0.8 | 314.5874 | 25442.9 | 8690.9 |
| 0.9 | 295.7002 | 32894.1 | 9726.7 |
| 1 | 263.4064 | 41585.9 | 10953.9 |

The graph of Power and delay with respect to voltage for 4-bit KSA is shown in the fig 13 and fig 14.

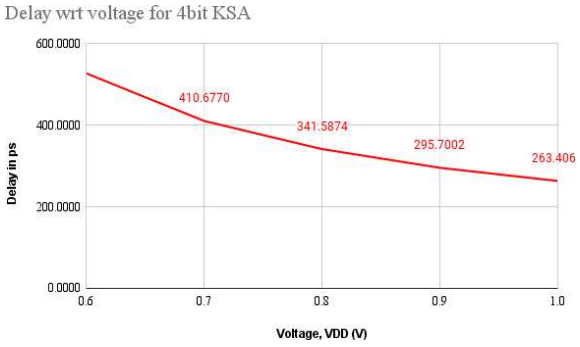



Fig. 13. Delay Vs Voltage: 4-bit KSA

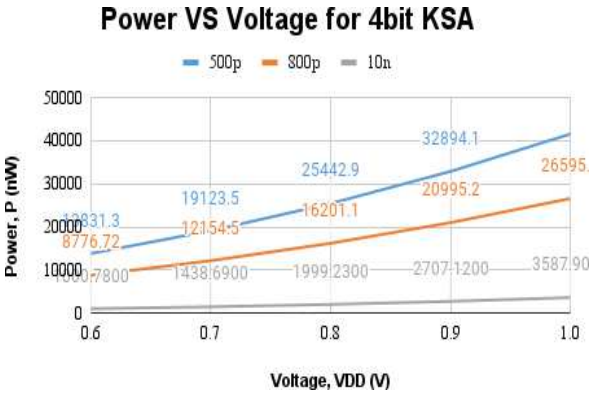



Fig. 14. Power Vs Voltage: 4-bit KSA

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TABLE VIII
8-BIT SIMULATIONS: AT 100MHZ

| VDD | Delay(ps) | Power at 100MHz(nW) | PDP(pJ) |
|---|---|---|---|
| 0.6 | 654.9645 | 2529.420 | 1656.6 |
| 0.7 | 510.9645 | 3617.480 | 1847.4 |
| 0.8 | 425.9626 | 4999.450 | 2129.5 |
| 0.9 | 369.2886 | 6729.410 | 2485.09 |
| 1 | 329.2833 | 8866.680 | 2919.6 |

TABLE IX
8-BIT SIMULATIONS: AT 1.25GHZ

| VDD | Delay(ps) | Power at 1.25GHz(nW) | PDP(pJ) |
|---|---|---|---|
| 0.6 | 654.9645 | 23048 | 150.9 |
| 0.7 | 510.9645 | 31891 | 162.8 |
| 0.8 | 425.9626 | 42444.5 | 180.7 |
| 0.9 | 369.2886 | 54862.7 | 202.6 |
| 1 | 329.2833 | 69287 | 228.1 |

In table 8-10, the Delay, Power, and Power-Delay Product (PDP) at various frequencies of 8-bit KSA are given. Figures 15 and 16 show a graph of power and delay as a function of voltage for an 8-bit KSA.

According to the graphs below, as the delay reduces, the power rises in relation to the voltage.

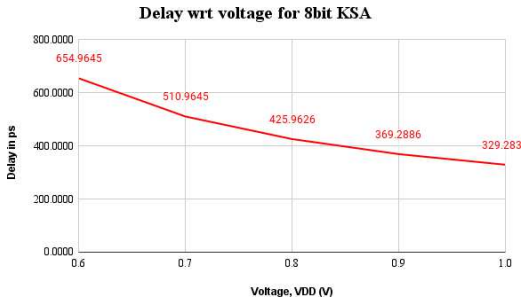



Fig. 15. Delay Vs Voltage: 8-bit KSA

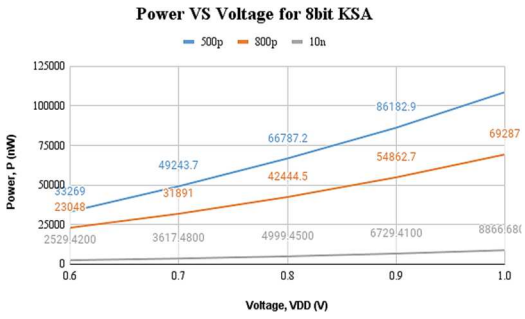



Fig. 16. Power Vs voltage: 8-bit KSA

The number of transistors used in 4-bit KSA is 240 and 8-bit KSA is 552 respectively.

The Area Utilization of all the sub-blocks, 4-bit and 8-bit KSA are tabulated which is shown in the table 11.

TABLE X
8-BIT SIMULATIONS: AT 2GHZ

| VDD | Delay(ps) | Power at 2GHz(nW) | PDP(pJ) |
|---|---|---|---|
| 0.6 | 654.9645 | 33269 | 217.9 |
| 0.7 | 510.9645 | 49243.7 | 251.4 |
| 0.8 | 425.9626 | 66787.2 | 284.4 |
| 0.9 | 369.2886 | 86182.9 | 318.2 |
| 1 | 329.2833 | 108625 | 357.6 |

TABLE XI
AREA UTILIZATION ($UM^2$)

| | Height(um) | Width(um) | Area($um^2$) |
|---|---|---|---|
| AND | 0.864 | 0.602 | 0.5201 |
| OR | 0.864 | 0.602 | 0.5201 |
| EX-OR | 0.864 | 0.688 | 0.5944 |
| Gen_Prop | 0.864 | 1.204 | 1.0403 |
| $\bar{Gi}_Pi$ | 0.864 | 1.376 | 1.1889 |
| $\bar{Gi}$ | 0.864 | 1.032 | 0.1889 |
| 4-bit KSA | 2.592 | 5.848 | 15.1580 |
| 8-bit KSA | 3.456 | 10.664 | 36.8548 |

## VI. CONCLUSION

The objective was to build the Kogge-Stone Adder, one among the fastest designs in its class. AND gates, OR gates, EX- OR gates, and Sub-Blocks like Gen prop Block, Gi Pi Block, and Gi Block are all made with the Standard Cells in the 12T track format. These Standard cells (Basic gates and Sub-Blocks) are used in KSA layout's Floor Planning.

The adder is designed in the 18nm technology node, and its schematics, symbols, layouts, and simulations are all used to examine its PPA parameters in their entirety, followed by the PDP, which is a major determining factor when comparing operation speed, and finally the transistor/Gate count of the adder architecture design.

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